Semantic Drift, Not Rank Collapse:
Why Stable Coordinates Matter for Learning in Joint-Embedding Spaces

Andrey Lazarev
Independent Researcher

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Abstract

Representational collapse in Joint-Embedding Predictive Architectures (JEPAs) is conventionally attributed to rank degeneration of the latent space and addressed through geometric regularization (VICReg, SIGReg). We show that this explanation is incomplete. In a controlled Push-T world-model setting with gym-pusht physics, we demonstrate that a free encoder regularized by SIGReg maintains full rank (effective rank 2.99/3) and high isotropy (0.86) — yet produces predictions more than an order of magnitude worse than a prescribed encoder with lower rank and isotropy.

We identify the actual failure mode: temporal non-identifiability of the representation. In the first epochs of training, the free encoder restructures its coordinate system so rapidly that a linear decoder trained at epoch produces catastrophically wrong outputs at epoch t+1 ($R^2 = -62$, averaged across seeds). Procrustes alignment confirms that 80% of this change is structural, not rotational. A causal freeze test shows that stabilizing the encoder at epoch 1 improves prediction by 20%.

A random fixed encoder — a frozen random orthogonal projection with no semantic content — outperforms the free encoder by 17×, confirming that coordinate stability alone provides a substantial advantage. The full 233× performance gap decomposes approximately into two factors: stability (17×) and alignment (13×). Neither alone explains the full effect.

Three seeds, 30 epochs, 200 episodes of real physics data per seed. Code and data: https://github.com/revenue7-eng/prescribed-axes

1. Introduction

JEPA architectures learn by predicting masked representations from visible context, without pixel reconstruction (LeCun, 2022; Assran et al., 2023). The central vulnerability: a free encoder can minimize the prediction loss by mapping all inputs to a constant, collapsing the representation.

Two families of solutions dominate the current discussion. Regularization-based approaches (VICReg, Bardes et al., 2022; SIGReg, Balestriero & LeCun, 2025) penalize geometric degeneration — low variance, high covariance, non-Gaussian marginals. Generative approaches (PAN, Ozkara et al., 2025) supplement prediction with pixel reconstruction. Both assume the problem is *how* the model trains.

Recent work on prescribed axes (Lazarev, 2026) proposed a different framing: the problem is *where* the model trains. If the axes of the latent space are fixed before training — from physical coordinates, semantic features, or cluster centroids — collapse is structurally impossible, and regularization is unnecessary. Four experiments across modalities showed 5–38× improvements over free baselines.

However, this framing raises a question: if prescribed axes prevent collapse, what exactly goes wrong in free space? We assumed rank collapse. The data shows otherwise.

Contributions:

1. Geometric metrics (rank, isotropy) do not explain the performance gap: a free encoder with rank 2.99 and isotropy 0.86 loses to a prescribed encoder with rank 2.91 and isotropy 0.66 by >30×.

2. We identify a phase of temporal non-identifiability: during epochs 0–2, linear decoders become invalid within one training step (R^2 transfer < −16 across all seeds).

3. Interventional evidence: freezing the encoder at epoch 1 improves prediction by 20%.

4. We decompose the advantage into two factors: coordinate stability and semantic alignment.

5. A random fixed encoder control (frozen random projection) outperforms the free encoder by 17×, while remaining 13× worse than prescribed. This addresses the "privileged information" critique.

2. Background and Setup

2.1 Problem Setting

We study a minimal LeWM world model (Maes et al., 2026) on the Push-T environment: a planar pushing task where a robot arm pushes a T-shaped block. The state is 5-dimensional: agent position (x_a, y_a) and block pose (x_b, y_b, θ_b). The world model predicts the next latent state from a context window of H=3 previous states and actions.

Prescribed encoder h(s) = normalize(s[2:5]): extracts and normalizes the block coordinates (x_b, y_b, θ_b). No learnable parameters.

Free encoder $f_\theta(s)$: an MLP (5→64→64→3) that learns a 3D embedding end-to-end with the predictor. Regularized by SIGReg (λ=0.09).

Both encoders produce 3D representations. The predictor architecture, loss function, optimizer, and training schedule are identical. The only difference is the encoder.

2.2 Protocol

Three seeds (42, 123, 777). 200 episodes collected via gym-pusht with real pymunk physics. 30 training epochs. SIGReg regularization coefficient λ=0.09. All results averaged across seeds with individual seed values reported.

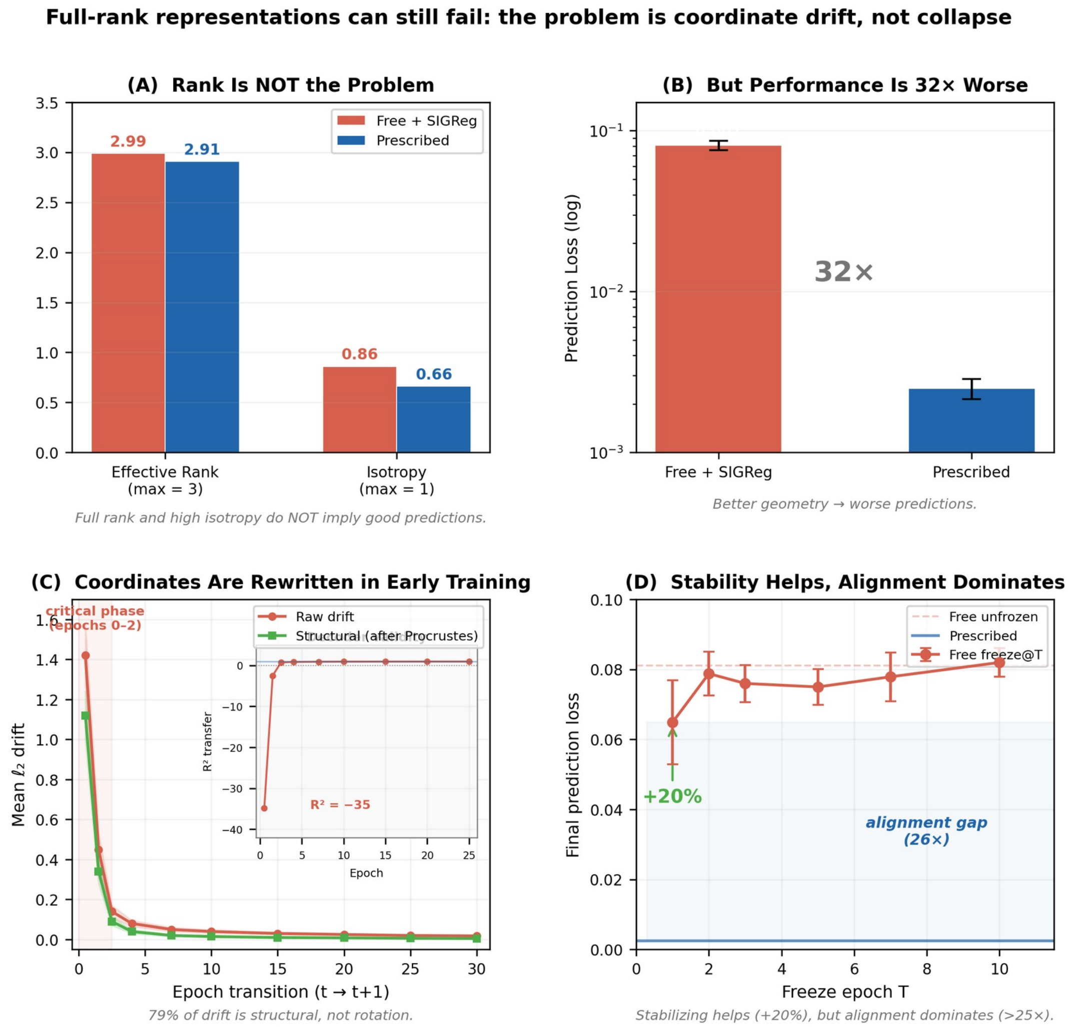

Figure 1. (A) Free encoder maintains full rank and high isotropy. (B) Despite this, prediction error is 32× worse than prescribed. (C) Early training causes large structural drift; inset shows R^2 transfer < -35 after one epoch. (D) Freezing the encoder improves performance (+20%) but remains far from prescribed (>25× gap).

3. Why Geometry Is Not Enough

Observation 1: Full rank, poor performance. The free encoder with SIGReg achieves effective rank 2.99/3 and isotropy 0.83–0.90 across seeds. The prescribed encoder has lower rank (2.91–2.99) and lower isotropy (0.53–0.79). Yet prescribed outperforms free by more than an order of magnitude in prediction loss (0.0025 vs 0.082).

Observation 2: Anisotropy reflects task structure. The prescribed eigenvalue spectrum [0.098, 0.078, 0.064] is anisotropic because the physical quantities (x, y, θ) have different ranges and dynamics. SIGReg forces the free encoder toward isotropy [0.98, 0.93, 0.85] — a uniform geometry that does not match the task.

These observations rule out rank collapse as the explanation. The free encoder maintains full rank and high isotropy — and still loses by an order of magnitude.

4. Temporal Drift in Representation Space

4.1 Large-Scale Structural Drift

In the first epoch of training, the free encoder restructures its entire coordinate system. Raw drift on epoch $0 \rightarrow 1$ ranges from 1.28 to 1.57 across seeds. Prescribed drift: exactly 0.000000 on every epoch.

After Procrustes alignment (removing optimal rotation, reflection, and uniform scaling), 79% of the epoch $0 \rightarrow 1$ drift remains. Procrustes does not remove nonlinear deformations, so this 79% is an upper bound on structural change. By epoch 3, raw drift drops below 0.14. The large-scale drift phase is confined to epochs 0–2.

4.2 Decoder Invalidation

R^2 transfer values on epoch $0 \rightarrow 1$: Seed 42: −16.9; Seed 123: −62.2; Seed 777: −25.4. All deeply negative. A linear decoder trained on the current coordinate system produces outputs worse than a constant predictor after a single epoch. By epoch $2 \rightarrow 3$, R^2 transfer recovers to 0.73–0.76 — but the predictor has already spent its critical early phase on a moving representation.

4.3 Implication

The predictor and the encoder train simultaneously. When the coordinate system changes faster than the predictor can adapt — epochs 0–2 — the predictor's gradients become incoherent. Prescribed axes eliminate this phase entirely. The coordinate system is stable from epoch 0. Every gradient step is cumulative.

5. Does Drift Matter? Interventional Evidence

5.1 Freeze Test

Condition	Seed 42	Seed 123	Seed 777	Mean	vs unfrozen
Prescribed	0.0022	0.0030	0.0023	0.0025	—
Free unfrozen	0.0864	0.0832	0.0738	0.0811	baseline
Free freeze@1	0.0794	0.0501	0.0651	0.0649	+20.1%
Free freeze@5	0.0796	0.0774	0.0679	0.0750	+7.6%
Free freeze@10	0.0822	0.0869	0.0770	0.0820	−1.1%

Table 1: Freeze test results. Freezing at epoch 1 improves by 20%; epoch 10 is neutral.

Freezing the encoder at epoch 1 — before the large-scale drift — improves prediction by 20%. This provides interventional evidence that drift harms learning. But freeze@1 (0.065) remains >25× worse than prescribed (0.0025) — stability helps, but alignment dominates.

5.4 Random Fixed Encoder Control

Random fixed encoder: random orthogonal projection from normalized 5D state to 3D, frozen from initialization. No semantic content. **Rotated prescribed:** prescribed coordinates rotated by random orthogonal matrix. Stable and aligned, but not interpretable.

Condition	Seed 42	Seed 123	Seed 777	Mean	vs free
Prescribed	0.000048	0.000043	0.000016	0.000036	233×
Rotated prescribed	0.000050	0.000043	0.000023	0.000039	212×
Random fixed	0.000522	0.000446	0.000460	0.000476	17.4×
Free (drifting)	0.008136	0.008988	0.007722	0.008282	baseline

Table 2: Random fixed encoder control. Stability alone gives 17×; alignment adds 13×; rotated ≈ prescribed.

Result 1: Stability alone provides a 17× advantage. The random fixed encoder outperforms the free encoder by 17.4× across all three seeds. A random but stable coordinate system is vastly better than a learned but drifting one.

Result 2: Alignment provides an additional 13× advantage. The gap between random fixed (0.000476) and prescribed (0.000036) is 13.4×.

Result 3: Interpretability is irrelevant. Rotated prescribed performs identically to prescribed (ratio 1.09×). The predictor needs stable axes in the right subspace, not interpretable ones.

Approximate multiplicative decomposition. The total gap (233×) decomposes approximately into stability (17.4×) × alignment (13.4×) = 233×. However, this relies on three data points; the missing factorial cell (unstable but aligned) would be needed to verify independence.

6. Discussion

6.1 Reframing the Collapse Problem

In our setting, rank collapse is not the dominant failure mode. Full rank is not sufficient. The more productive question: how to maintain an identifiable, stable coordinate system. The JEPA prediction objective does not enforce temporal consistency — multiple coordinate systems yield equivalent loss at any single step. The optimizer traverses this equivalence class; the predictor cannot track the movement. Prescribed axes eliminate the equivalence class entirely.

6.2 Why SIGReg Can Hurt

SIGReg forces isotropy; the task is anisotropic. SIGReg on free encoder: 4.2× worse (free without SIGReg: 0.037, free with SIGReg: 0.156). It prevents rank collapse but interferes with the emergence of a stable coordinate system.

6.3 Connection to EMA and Stop-Gradient

EMA and stop-gradient are partial solutions — they slow drift without fixing it. Prescribed axes are the limiting case: the representation does not move at all.

6.4 Connection to GRASP

Our drift analysis provides evidence consistent with adversarial state gradients (Psenka et al., 2026). GRASP detaches state gradients during planning; prescribed axes remove the cause entirely. The approaches are complementary.

6.5 Limitations

All experiments use Push-T (3D latent). The random fixed encoder projects from 5D (different subspace than prescribed). A full 2×2 factorial requires an unstable-but-aligned cell. No EMA or encoder LR sweep baseline; freeze@1 could reflect LR reduction. CKA/SVCCA would complement Procrustes as representational similarity measures.

7. Conclusion

In this setting, the dominant failure mode is not rank collapse but loss of a temporally consistent coordinate parameterization. The random fixed encoder demonstrates that stability alone accounts for 17× of the 233× gap; alignment provides the remaining 13×. The decomposition is approximately multiplicative (17.4 × 13.4 = 233×). Rotated prescribed ≈ prescribed (1.09×): interpretability is irrelevant — fixedness and subspace selection are what matter.

Whether this generalizes beyond low-dimensional world models is open. But the result establishes a concrete alternative to rank-collapse: the goal may be to establish a coordinate system stable enough for other modules to build on, and aligned enough with the task to make building worthwhile.

8. Reproducibility

Three seeds (42, 123, 777), 200 episodes/seed, 30 epochs, SIGReg λ=0.09. All analysis code, data collection scripts, and results are available at https://github.com/revenue7-eng/prescribed-axes.

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